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Abstract: This TD merges C310, C369 and C385 with modification and updates the baseline text of [XSTR.IMT2030-sec-con](#) based, which is the outcome of SG17 in December 2025 SG17 meeting.

Draft new Technical Report ITU-T XSTR.IMT2030-sec-con

Security Consideration for IMT-2030 Networks

Summary

This Technical Report describes the progress of standardization of International Mobile Telecommunications (IMT) for 2030 and beyond (IMT-2030), analyzes the key driving force of security of IMT-2030 from the trends of IMT-2030, and analyzes the potential security challenges and issues based on the scenarios and capabilities of IMT-2030. At last, it gives considerations of the standardization work for security of IMT-2030.

This Technical Report can help the stakeholders figure out what factors will impact security considerations and what challenges and issues need to be dealt with facing IMT-2030. It will be helpful for further study in the future on formulating the detailed threats, requirements, design and solutions of IMT-2030 security. This Technical Report can be used as a guideline and reference for the further development of IMT-2030 networks security related documents.

Keywords

IMT-2030, security, resilience, challenges

Change Log

This document contains Version 3 of the ITU-T Technical Report *XSTR.IMT2030-sec-con on "Security Consideration for IMT-2030 Networks" approved at the ITU-T Study Group SG 17 meeting held in Geneva, December 2025.

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서식 지정함: 프랑스어 (프랑스)

서식 지정함: 프랑스어 (프랑스)

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Security considerations for IMT-2030 networks

1. Scope

On the basis of the progress made by ITU on International Mobile Telecommunications (IMT) for 2030 and beyond (IMT-2030), including future trends and scenarios and overall objectives [ITU-R M.2160-0], this Technical Report describes the following considerations on:

- The key driving force of security of IMT-2030.
- The security challenges and issues based on the use cases and capabilities of IMT-2030.
- The standardization work for security of IMT-2030.

2. References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Technical Report. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Technical Report are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published.

The reference to a document within this Technical Report does not give it, as a stand-alone document, the status of a Recommendation.

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|----------------------|--|
| [ITU-T Y.3090] | Recommendation ITU-T Y.3090 (2022), <i>Digital twin network – Requirements and architecture</i> . |
| [ITU-T TR.terms] | ITU-T TR.IMT2030-terms (2024), <i>IMT-2030 networks - Considerations on terms and definitions</i> . |
| [ITU-R M.2160-0] | Recommendation ITU-R M.2160-0 (2023), <i>Framework and overall objectives of the future development of IMT for 2030 and beyond</i> . |
| [ITU-R M.2516-0] | Recommendation ITU-R M.2516-0 (2022), <i>Future technology trends of terrestrial International Mobile Telecommunications systems towards 2030 and beyond</i> . |
| [ITU-T X.2011] | Recommendation ITU-T X.2011 (2024), <i>Security guidelines for digital twin network</i> . |
| [ITU-T X.dtns] | Recommendation ITU-T Y.dtns (2025), <i>Guidelines of using digital twin of network for network security</i> . |
| [ITU-T TR.5Gsec-bsf] | Recommendation ITU-T TR.5Gsec-bsf (2025), <i>Guidelines for a telecommunication network built-in security framework</i> . |
| [3GPP TR 22.870] | Study on 6G Use Cases and Service Requirements: Stage 1 |

3. Definitions

3.1. Terms defined elsewhere

This Technical Report uses the following terms defined elsewhere:

3.1.1 security [ITU-R M.2160-0]: preservation of confidentiality, integrity, and availability of information, such as user data and signalling, and protection of networks, devices and systems against cyberattacks such as hacking, distributed denial of service, man in the middle attacks, etc.

3.1.2 digital twin network [ITU-T Y.3090]: A virtual representation of a physical network. It is useful for analysing, diagnosing, emulating and controlling the physical network based on data, model and interface, to achieve the real-time interactive mapping between the physical network and virtual twin network.

3.1.3 IMT-2020 [ITU-T Y.3100]: (Based on [b-ITU-R M.2083-0]) Systems, system components, and related technologies that provide far more enhanced capabilities than those described in [b-ITU-R M.1645].

3.1.4 dedicated network [ITU-T Y.3114]: A network designed for application domains with common requirements.

3.1.5 resilience [ITU-R M.2160-0]: capabilities of the networks and systems to continue operating correctly during and after a natural or man-made disturbance, such as the loss of primary source of power, etc.

3.2. Terms defined in this Technical Report

This Technical Report defines the following terms:

3.2.1 non-terrestrial network: communication networks that are not based on traditional terrestrial infrastructure (like cell towers) such as satellites, unmanned aerial vehicles.

4. Abbreviations and acronyms

This Technical Report uses the following abbreviations and acronyms:

AI	Artificial intelligence
XR	Extended Reality
IMT-2030	International Mobile Telecommunications-2030

5. Conventions

None.

6. Analysis on key driving force of security of IMT-2030

[Editor note: This chapter analyze the key driving force of security of IMT-2030 based on the listed trend in ITU-R M.2160-0. Here are some examples. More detailed analysis will be provided in the future version]

- **Motivation and societal considerations** [ITU-R M.2160-0]: The motivation for the development of IMT-2030 is to continue to build an inclusive information society towards contributing to support the United Nations Sustainable Development Goals (SDGs). To this

end, IMT-2030 is expected to be an important enabler for achieving the goals, such as inclusivity, ubiquitous connectivity, sustainability, innovation, enhanced security and resilience, standardization and interoperability, interworking. For **Enhanced security and resilience**, *IMT-2030 system is expected to be secure by design. It is expected to have the ability to continue operating during and quickly recover from a disruptive event, whether natural or man-made. Making security and resilience as the key considerations in the design, deployment and operation of IMT-2030 systems is fundamental to achieving broader societal and economic goals.* Mobile communication is a key infrastructure to support modern society and IMT-2030 is expected to drive the next wave of digital economic growth, as well as sustainable far-reaching societal changes, digital equality and universal connectivity. Firstly, IMT-2030 is expected to foster **innovation** to facilitate connectivity, productivity and the efficient management of resources, which will also drive the security in a direction of innovation. IMT-2030 is also expected to include affordable and meaningful connectivity between everyone and a wide range of services (i.e., **Inclusivity** goal and **Ubiquitous connectivity** goal) and be built on energy efficiency, low power consumption technologies, reducing greenhouse gas emissions and appropriate use of resources (i.e., **Sustainability** goal). The inclusivity goal need consideration on public safety and data explosion related security issues like data confidentiality and privacy. The sustainability goal needs consideration on the balance between security effect and security cost (e.g., Lightweight and moderate security), and on the balance between security redundancy and network simplification in specific scenarios.

- **User and application trends:** Future applications and services enabled by IMT-2030 are expected to connect humans, machines and various other things together, and work with a digital world that extends the real world, such as immersive experience, telepresence, multimodal interaction, sensing, extended reality (XR) and artificial intelligence (AI). So IMT-2030 is expected to integrate sensing and AI-related capabilities into communication and serve as a fundamental infrastructure to enable such new user and application trends. The enhanced service value of the IMT-2030 in the future is undoubtedly very attractive to profit-driven attackers. IMT-2030 is expected to further enhance security and resilience. Stronger, fine-grained and flexible on-demand security protection is needed to ensure the dynamic adaptation of the network, the user, application and business scenarios.
- **Technology trends:** Technologies to enhance the radio interface and network are in rapid development. These emerging technologies might not only improve the network capabilities, but also pose more severe challenges for the security of IMT-2030 which will become one of the key driving forces of security evolution. Firstly, the security risks of new technology itself when using new technology in the network need to be paid attention to like AI data poisoning; Secondly, the development of some technologies will bring some challenges to the security of the existing communication system. For example, quantum computing technology brings risks to the existing cryptosystem [ITU-T X.1811]; Thirdly, the development of these new technologies/principles might be able to promote the effectiveness of security protection and privacy. It is necessary to consider how to introduce and apply these new technologies in IMT-2030 to improve the level of security protection, such as trusted computing, blockchain, zero trust, etc.
- **Other trends:** There can also be other trends like network architecture trend which might become the driving forces of security evolution. So it is necessary to consider and analyze which trends and factors can and how to impact the security issues and security considerations surrounding IMT-2030. It will be helpful for considering and formulating the threats, requirements, design and solutions of IMT-2030 security in the future.

7. Analysis on security challenges for usage scenarios and capabilities of IMT-2030

[Editor note: This chapter analyzes the security challenges and issues based on the use cases and capabilities of IMT-2030. Here are some examples. More detailed analysis on more use cases will be provided in the future version.]

IMT-2030 is expected to expand and support various user, application and technology trends, while providing prospects towards a sustainable digital transformation. IMT-2030 is expected to be built on overarching aspects which act as design principles commonly applicable to all usage scenarios. Usage scenarios of IMT-2030 are envisaged to include not only expanded IMT-2020 usage scenarios, but also new usage scenarios arising from capabilities, such as artificial intelligence and sensing. In this Technical Report, we will analyze the potential security challenges of the use cases included in 6 kinds of usage scenarios and capabilities of IMT-2030 listed in ITU-R M.2160. This can help the stakeholders to figure out the security issues IMT-2030 need to tackle with. Here are some security challenge examples for the 6 usage scenarios and related capabilities.

7.1 Security challenges and key issues for Immersive Communication

This usage scenario extends the enhanced Mobile Broadband (eMBB) of IMT-2020 and covers use cases which provide a rich and interactive video (immersive) experience to users, including the interactions with machine interfaces. Typical use cases include communication for immersive XR, remote multi-sensory telepresence, and holographic communications. IMT-2030 will put forward higher requirements for these massive, high-dimensional data communication rates. 300 Mbit/s and 500 Mbit/s are given as possible examples for the user experience rate and 50, 100, 200 Gbit/s are given as possible examples for peak data rate [ITU-R M.2160-0].

7.1.1 Security challenges brought by extremely high-data-rates

[Challenge IC-1-1] In order to match the transmission performance of ultra-high rate data stream, it is a key issue to realize the encryption and integrity protection of ultra-high data rate stream, and the difficulty of matching the key rate and the communication rate is the key problem that restricts its security ability, and the traditional security means to improve the key rate will occupy a lot of communication resources.

[Challenge IC-1-2] IMT2030/6G will enable massive and densely connected ecosystems of IoT, XR, haptic, and autonomous devices, many of which operate under strict power and processing constraints. These limitations, combined with the extremely high data flows in 6G networks, make centralized security orchestration—such as cloud-based SOC—impractical, necessitating lightweight and distributed security mechanisms closer to the network edge.

[Challenge IC-1-3] Ultra-high throughput in IMT2030/6G networks enables the rapid collection and dissemination of vast volumes of behavioral, sensory, and contextual data generated by massive numbers of IoT devices—including wearable and in-body health sensors—within XR and tactile Internet environments.

7.1.2 Security challenges brought by multi-modal sensory data

[Challenge IC-2-1] These use cases introduce more special human-centric multi-modal sensory data transferred including human's audio, video, taste, odour, haptic and emotion information. Due to the irrevocable nature of human biometric data, it represents a very important privacy issue to be tackled by IMT-2030.

7.1.3 Security challenges brought by connection among virtual objects

[Challenge IC-3-1] Immersive Communication connects real and virtual objects. The user may have different identities in the virtual world, and real world, and physical world. [TR 22.870]

[Editor note: clarify the difference of real world, and physical world, or else just remain one]

Virtual objects like digital humans need to have its own digital identity associated with the user to ensure their trusted access to the network. For an embodied robot, there is a need to ensure the association with a controller/user for its behaviour control. How to build more flexible and secure identity identification and trust relationships among multiple parties is an issue to meet the needs of users to switch and interact in different virtual environments.

7.2 Security challenges and key issues for Hyper Reliable and Low-Latency Communication

This usage scenario extends the Ultra-Reliable and Low-Latency Communication (URLLC) of IMT-2020 and covers specialized use cases that are expected to have more stringent requirements on reliability and latency including communications in an industrial environment for full automation, control and operation such as machine interactions, emergency services, tele-medicine, and monitoring for electrical power transmission and distribution. The research target of latency (over the air interface) could be 0.1 – 1 ms for IMT-2030 [ITU-R M.2160-0]. This capability is essential for mission-critical use cases, such as remote surgery, autonomous driving, industrial automation, UAV swarm control, and smart grid management.

7.2.1 Security challenge brought by Ultra-Reliable

[Challenge HRLL-1-1] It is a key issue to consider Ultra-Reliable which means that the network can still maintain high availability and high reliability in the face of attackers.

[Challenge HRLL-1-2] For the hyper reliable redundant transmission, it is necessary to consider that the redundant path and the single path have the same security level.

[Challenge HRLL-1-3] Unavailability of network such as Denial of Service (DoS) becomes Catastrophic as even tiny disruptions are unacceptable.

[Challenge HRLL-1-4] Violation of data integrity becomes critical as a small alteration of data can cause dangerous behaviours (e.g., reversing the direction of a drone).

7.2.2 Security challenge brought by Low-Latency

[Challenge HRLL-2-1] Low-Latency means that the network needs to identify, protect, detect, response and recovery from a disruptive event in a rapid way.

Latency refers to the contribution by the network to the time from when the source sends a packet of a certain size to when the destination receives it. Low-latency means that the said time is short.

[Challenge HRLL-2-12] Traditional communication security is usually implemented crossing multiple protocol layers and physical domains on the terminals and network, which might lead to large delay of security info transmission and processing. Low-Latency means that the security mechanisms (e.g., encryption, authentication) require limited computation time, which leads to themeas network nodes often operate with less robust physical and logical security simplification of security mechanisms, resulting in weakened protection capabilities or an accelerated spread of cyberattacks, exposing endpoints to attacks such as tampering, spoofing, and side-channel attacks. Low-Latency also means that intrusion detection, anomaly detection, and access verification may not have sufficient time to identify or mitigate threats, increasing the risk that real-time attacks could impact safety-critical operations. There might be a contradiction between low latency and high security. So it is a key issue to consider more efficient security mechanisms to ensure hyper reliable communication.

~~[Challenge HRL-2-3] Low Latency means that malicious traffic or malware can spread faster across the network.~~

~~[Challenge HRL-2-3] Low Latency means that AI based intrusion detection systems (IDS) need reduced time to analyse behaviours for Intrusion Detection and Response.~~

~~[Editor note: "Latency over the air interface refers to the contribution by the radio network to the time from when the source sends a packet of a certain size to when the destination receives it." Low-latency means that the said time is short. The exact intention of the current text above need better clarity.]~~

~~For example, traditional communication security is usually implemented crossing multiple protocol layers and physical domains on the terminals and network, which might lead to large delay of security info transmission and processing. There might be a contradiction between low latency and high security. So it is a key issue to consider more efficient security mechanisms to ensure hyper-reliable communication.~~

서식 지정함: 글꼴: 기움임꼴 없음

7.3 Security challenges and key issues for Massive Communication

This usage scenario extends massive Machine Type Communication (mMTC) of IMT-2020 and involves connection of massive number of devices or sensors for a wide range of use cases and applications. Typical use cases include expanded and new applications in smart cities, transportation, logistics, health, energy, environmental monitoring, agriculture, and many other areas such as those requiring a variety of IoT devices without battery or with long-life batteries. This usage scenario would require support of high connection density which could be $10^6 - 10^8$ devices/km² for IMT-2030. [ITU-R M.2160-0]

[Challenge mMTC-13-1] Authenticating billions of low-power, resource-constrained IoT devices without overloading the network or rapidly depleting device batteries poses a significant challenge.

[Challenge mMTC-13-2] Many devices with limited processing power and memory may resort to weak or outdated encryption to conserve resources, as traditional cryptography is often too heavy and energy-intensive, thereby risking exposure of sensitive data such as healthcare or industrial sensor information.

[Challenge mMTC-13-3] A massive number of devices can be exploited to generate excessive signalling requests, overwhelming the control plane and causing network degradation or even denial of service.

[Challenge mMTC-13-4] dedicated network slices for specific IoT services can allow an attack on one slice to spread if isolation is insufficient.

7.3.1 Security challenge brought by high connection density

[Challenge MC-21-1] Large number of concurrent requests from devices might trigger signaling storm and services fail. It is a key issue to consider the prevention of distributed denial of service (DDoS) attacks from devices.

[Challenge MC-21-2] Traditional encryption and access authentication protocols might have problems such as massive encryption and difficult management of authentication key distribution on the network side, and high complexity and insufficient security intensity on the terminal side. It is a key issue to design encryption and access authentication protocols which are suitable for high connection density.

7.3.2 Security challenge brought by devices without battery

[Challenge MC-32-1] Devices without battery often have limited computing and storage resources. This characteristic makes it difficult for these devices to be equipped with complex encryption and protection mechanisms. At the same time, they rely on environmental energy for power supply, which is unstable and limited. This makes them vulnerable to energy depletion attacks, causing the devices to stop working prematurely and affecting business operations.

7.4 Security challenges and key issues for Ubiquitous Connectivity

This usage scenario is intended to enhance connectivity with the aim to bridge the digital divide. Connectivity could be enhanced through interworking with non-terrestrial networks of IMT like satellite communication systems, as well as with other non-IMT terrestrial networks like broadcast [ITU-R M.2160-0].

7.4.1 Security challenge brought by non-terrestrial communication systems

[Challenge UC-1-1] The future network integrates non-terrestrial networks, and non-terrestrial networks and ground mobile networks adopt different standard systems. Long-range, space-based links may be susceptible to jamming, spoofing, or eavesdropping. Heterogeneous integration poses security risks of compatibility and interoperability. It is a key issue to consider how to ensure the security of integration of the different networks.

[Challenge UC-1-2] Space-ground integrated network has the characteristics of open communication environment, time-varying communication node topology, complex network structure, and low processing ability of space-based nodes, which increases the risk of the network and makes the potential vulnerability easier to be exploited by attackers.

7.4.2 Security challenge brought by heterogeneous network

Ubiquitous connectivity and wide area coverage introduce the integration and interworking of heterogeneous network and diversified terminal environment

[Challenge UC-2-1] Integration of heterogeneous networks brings in wide-area scheduling and orchestration issues of security strategy. Different security requirements for each distributed and heterogeneous networks need to be met. The consistency and continuity of security protection and networks need to be ensured when users or services switch and interact among different networks.

[Challenge UC-2-2] Interworking of heterogeneous networks brings in cross-domain issues. Frequent cross-domain authentication brings about network latency and management cost, as well as threats of single point of failure. What's more, it is difficult to establish a common root of trust for cross domain communication. Security isolation of these networks is crucial to avoid cross-attack risks. Fine-grained security isolation based on the security needs of different businesses requirements is needed.

7.5 Security challenges and key issues for Artificial Intelligence and Communication

This usage scenario would support distributed computing and AI applications. Typical use cases include IMT-2030 assisted automated driving, autonomous collaboration between devices for medical assistance applications, offloading of heavy computation operations across devices and networks, creation of and prediction with digital twins, and others. This usage scenario is expected to include a set of new capabilities related to the integration of AI and compute functionalities into IMT-2030, including data acquisition, preparation and processing from different sources, distributed AI model training, model sharing and distributed inference across IMT systems, and computing resource orchestration and chaining [ITU-R M.2160-0].

7.5.1 Security challenges brought by Artificial intelligence

[Challenge AIC-1-1] AI/ML will play a critical role in IMT-2030 for network automation, threat detection, and optimization. However, AI models are susceptible to adversarial attacks such as evasion, model poisoning, model inversion, and membership inference. Adversarial attacks degrade, manipulate, or mislead ML models by exploiting their vulnerabilities. The model poisoning attack corrupts a model during training by injecting malicious data, influencing its predictions or causing targeted misclassifications. The model inversion attack extract or infer sensitive information (e.g., training data or private attributes) from a trained model. There are security threats introduced in the process of using AI. For example, AI faces security threats such as poisoning or unbalanced training data and algorithm bias discrimination and so on.[XSTR-SEC-AI - Guidelines for security management of using artificial intelligence technology]. Digital twins face security challenges such as untrusted twin data, unreliable control instructions, and insecure twin models [ITU-T X.2011]. As AI and digital twin are expected to be applied to help decision on the network operation and prediction of the network status, the threats faced might cause improper AI-driven or digital-twin-based network decisions and need to be addressed.

[Challenge AIC-1-2] AI and digital twin technology can be leveraged to automate detection and disposal of new threats, anomalies and cyberattacks in IMT-2030. For example, through multi-dimensional data mining and learning, AI can model or extract features of network attack behavior and threat intelligence, identify known or unknown malware related to network assets, and adaptively orchestrate optimal set of security strategies and security network devices. It's a challenge to consider how to use and integrate AI and digital twin for network security in IMT-2030.

7.5.2 Security challenges brought by digital twin

With real-time interactive mapping between the physical and virtual twin networks, digital twin networks (DTNs) can help efficiently and intelligently verify, simulate, deploy and manage IMT-2030 networks [ITU-R M.2160-0]. Security challenges brought by digital twin include the security threats of using DTN in IMT-2030 and the potential to use DTN to enhance security of IMT-2030.

[Challenge AIC-2-1] Digital twins face security threats such as untrusted twin data, unreliable control instructions, and insecure twin models [ITU-T X.2011]. Digital twins are expected to be applied to help decision on the network operation and prediction of the network status. The threats faced might cause improper digital-twin-based network decisions and need to be addressed.

[Challenge AIC-2-2] With the profound and full-scale understanding of historical, real-time and static network related data in digital twin network and the twin network as a test bed, security risk or security devices could be simulated and different network security strategies could be tested, optimized and decided [ITU-T X.dtns]. It is a challenge to consider how to use and integrate digital twin for network security in IMT-2030.

7.6 Security challenges and key issues for Integrated Sensing and Communication

This usage scenario makes use of IMT-2030 to offer wide area multi-dimensional sensing that provides spatial information about unconnected objects as well as connected devices and their movements and surroundings. Typical use cases include IMT-2030 assisted navigation, activity detection and movement tracking (e.g. posture/gesture recognition, fall detection, vehicle/pedestrian detection), environmental monitoring (e.g. rain/pollution detection), and provision of sensing data/information on surroundings for AI, XR and digital twin applications. Along with the provided communication capabilities, this usage scenario requires support of high-precision positioning and sensing-related capabilities, including range/velocity/angle estimation, object and presence detection, localization, imaging and mapping.

[Challenge ISC-1-1] Integrated Sensing and Communication makes the wireless environment of IMT-2030 system more vulnerable to security threats, such as eavesdropping of communication information, deceptive jamming attacks, denial of service (DOS) of communication functions, and privacy leakage of sensory signals. For example, heterogeneous sensor nodes have limited resources, which are easy to be hijacked and cause malicious damage to the system perception and communication network, and reduce the performance of the perception function, data processing function or others. The precise positioning capability provided by sensing technology especially requires the protection of user privacy and business secrets of enterprise information.

7.7 Other security challenges and key issues

7.7.1 Security challenges brought by quantum computing technology

[Challenge o-pqc-1] Quantum computing poses a threat to classical cryptographic algorithms (e.g., RSA, ECC) used in IMT-2030 networks. Future quantum attacks could break encryption, exposing sensitive data. The security strength of a symmetric cryptographic algorithm will be reduced to half due to Grover's quantum algorithm. Many commonly used asymmetric cryptographic algorithms, such as RSA, DSA, and elliptic curve-based algorithms, will provide no security due to Shor's quantum algorithm. Authentication, encryption and integrity will still be the main security functions in IMT-2030 networks, all of which need the support of cryptography technology. Among them, public key cryptography is used in key sharing, authentication and encryption, and they face more severe security threats from quantum computing. It is a key issue to develop post quantum cryptography algorithms which can resist quantum computing attacks, identify the cryptographic usage in IMT-2030 networks, and influenced network devices/modules which may need post quantum cryptography (PQC) migration. Gradual migration from current classical cryptographic systems to post quantum cryptographic system poses compatibility and interoperability risks.

[Challenge o-pqc-2] There are differences in performance among different categories of post-quantum cryptographic algorithms. Post-quantum cryptographic algorithms with larger key or ciphertext sizes may require the expansion of the length of protocol fields. At the same time, the increase in the scale of keys and ciphertext also poses higher requirements for storage space and transmission bandwidth, further incurring overheads such as keys' distribution delay, ciphertext's transmission delay, and processing delay. Algorithms with slow key generation speed and slow computing speed will increase the execution time of security services, including the time for connection establishment, which may affect the efficiency of business services. Especially for some latency-sensitive services, the service availability may be affected. IMT-2030 networks will have a wide variety of types, a large number of devices, and complex business functions, which brings stringent requirements for performance and stability. It is a key issue to identify the cryptographic usage in IMT-2030 networks, and influenced network devices/modules, and then select applicable algorithms according to the characteristics of the scenarios. It is a challenge to address issues such as insufficient performance support, high integration difficulty, and large cost overheads when applying post-quantum cryptographic algorithms.

[Challenge o-pqc-3] Breakthroughs in cryptanalysis and increased computing power can weaken the strength of cryptographic algorithms. If the IMT-2030 networks lack crypto-agility, algorithm replacement from weak cryptographic algorithms to strong ones can be extremely disruptive and often takes decades to complete [b-NIST-CSWP]. It is a key issue for the IMT-2030 networks to have the ability to migrate easily from one cryptographic algorithm to another in a way that is flexible, scalable, and dynamic for future smooth migration.

7.7.2 Security challenges brought by resilience

Several factors may contribute to the root cause of network failures [3GPP TR 22.870]:

- System unavailability may occur due to factors such as transmission connection loss or large-scale network outages.
- Network attacks, including distributed denial-of-service (DDoS) and other external threats, can undermine both the integrity and the availability of critical services.
- Force majeure events—such as typhoons, earthquakes, or other uncontrollable natural disasters—pose significant risks to network stability.
- Network operational errors, including misconfigurations, improper maintenance activities, or faults during software upgrades and commissioning, may also disrupt the normal operation of 6G systems.

[Challenge o-r-1] Network accidents caused by network attacks, network operational errors or force majeure events might lead to network outages causing disruptions in applications and affecting a vast number of users. IMT-2030 is expected to further enhance security and resilience which refers to capabilities of the networks and systems to have the ability to continue operating during and quickly recover from a disruptive event, whether natural or man-made [ITU-R M.2160-0]. It is a key issue to consider how to detect, respond, and recover from network accidents in a rapid way.

~~[Challenge o-r-2] One major security challenge in 6G networks arises from the massive attack surface created by hyper-connectivity. With billions of interconnected devices—including IoT nodes, autonomous vehicles, drones, AR/VR platforms, and industrial systems—each endpoint represents a potential entry vector for attackers. Ensuring service resilience in the face of large-scale botnet-driven DDoS attacks therefore becomes a critical concern.~~

[Challenge o-r-23] Another critical security challenge in 6G networks lies in ensuring supply chain and hardware integrity. Since 6G infrastructure will rely on globally sourced components such as chips, antennas, and sensors, the presence of hardware Trojans or hidden backdoors at the supply chain level could compromise the core infrastructure and severely undermine network resilience.

7.7.3 Security challenges brought by framework design

Facing the significant increase in security requirements and complexity brought by future network evolution, it is necessary to consider the problem from the perspective of the overall network: How to conduct security design to achieve a triangular balance between network quality, capability, and efficiency [ITU-T TR.5Gsec-bsf].

[Challenge o-fd-1] New requirements often require the introduction of new security capabilities, and with the evolution of the network, the increase of the network's processing chain and the continuous addition of security detection and protection mechanisms will inevitably lead to a decline in overall network performance. Therefore, it is necessary to consider the issue of how to avoid performance degradation caused by the continuous addition of security capabilities, and balance security with network performance and functional requirements.

[Challenge o-fd-2] Security design often lags behind, and is limited by, communication network design. Most measures can only protect the network through plug-ins or patches, which affect the performance, efficiency and security of the communication. Firstly, security disposal outside facilities and processes may incur additional communication costs and transmission risks. Some security functions may require invoking internal data of network elements and are more suitable for local execution. Secondly, the centralized security disposal may become a bottleneck point, which may affect the performance of the network flow passing through the disposal point and the robustness of security disposal. Finally, the evolution of security function can only be carried out in a patched manner, making it difficult to implement overall and active protection measures which limit the security protection effect. Furthermore, it is also not possible to fully utilize the security potential of the network itself. Therefore, it is necessary to consider how to avoid the separation of security architecture and network architecture, and achieve an integrated architecture.

[Challenge o-fd-3] As future networks become dynamically variable, and users, services, and networks brings diverse security requirements. These requirements pose significant security challenges to network management and operation, resource allocation and utilization, as well as data interaction and transmission. It is difficult to design a one-fit-all security solution, which gives rise to a key issue: developing an architecture with differentiation, self-adaptation, and scalability to dynamically adapt to heterogeneous networks, diverse terminals, and complex business scenarios, thereby ensuring business continuity and security.

7.7.4 Security challenges brought by supply chain

[Challenge o-sc-1] Software supply chain security refers to the practice of protecting the integrity and security of all software components involved in developing and deploying IMT-2030 network infrastructure, including the code, tools, libraries, and processes used throughout the development lifecycle, by implementing security measures at every stage to prevent vulnerabilities from being introduced and exploited by malicious actors, especially considering the complex and interconnected nature of IMT-2030 networks with a vast array of devices and potential attack vectors. It is a key issue to consider how to achieve the software supply chain security.

7.7.5 Security challenges brought by the measurable security indicators and evaluation environment

[Challenge o-si-1] The security performance of IMT systems has not yet been defined by quantitative indicators. In the current IMT-2030 framework, security is designated as a qualitative item rather than a measurable performance parameter. The absence of unified measurement indicators may lead to inconsistent security levels among operators, which can result in a decline of end-to-end network trustworthiness. Therefore, it is necessary to consider how to define measurable quantitative indicators for evaluating the security performance of communication systems.

[Challenge o-si-2] Current security evaluations rely on different criteria and methodologies depending on service scenarios and operators, resulting in inconsistent and incomparable outcomes. Due to the absence of measurement methods and evaluation environments for objectively assessing security performance, such as encryption strength, intrusion detection accuracy, recovery time, and trust levels, it is difficult to validate or certify the effectiveness and efficiency of IMT-2030 security functions under diverse network conditions. Therefore, it is a key issue to establish standardized methodologies and reproducible evaluation environments that can assess security performance under various network conditions and threat models.

//editor note: The new issues need to be mapped to figure 1.

7.8. Considerations of standardization work for security of IMT-2030

[Editor note: This chapter gives considerations on what standardization work can be considered to carry out in the future and the timeline based on the overall timeline

Based on the different attributes of standards, 6G security standards are classified from two dimensions:

● Dimension of Standard Research Objects

Considering the research objects involved in communication networks and security, 6G security standards can be divided into terminal security standards, network infrastructure security standards, network function security standards, data security standards, application/service security standards, etc.

1. Terminal security standards study the security of multi-form terminals that access 6G networks and adapt to 6G scenario-based services.

서식 지정함: 글꼴: 기움임꼴 없음

서식 있음: 들여쓰기: 첫 줄: 0 글자

서식 있음: 첫 줄: 0 글자, 글머리 기호 + 수준: 1 + 맞춤
위치: 0 cm + 들여쓰기 위치: 0.74 cm

서식 있음: 글머리 기호 및 번호 매기기

서식 있음: 들여쓰기: 첫 줄: 0 글자

2. Network infrastructure security standards address dedicated security equipment and the security of network infrastructure (e.g., storage resources, connectivity resources and computing resources) supporting 6G network functions.

3. Network function security standards addresses the security of network functions and the implementation of security through network functions which are used for network transmission, control, services, management and operation, etc.

4. Data security standards focus on the security of various types of data involved in 6G networks to ensure full-lifecycle data security.

5. Application/Scenario security standards develop customized security technology standards for different services and application scenarios while ensuring the security of 6G networks themselves, so as to meet scenario-specific characteristics and requirements which are listed in ITU-R M.2160.

6. General and Fundamental Technology provides fundamental support for 6G security including standards on roadmap, security architecture and those on common techniques for multiple objects.

● Dimension of Standard Stages

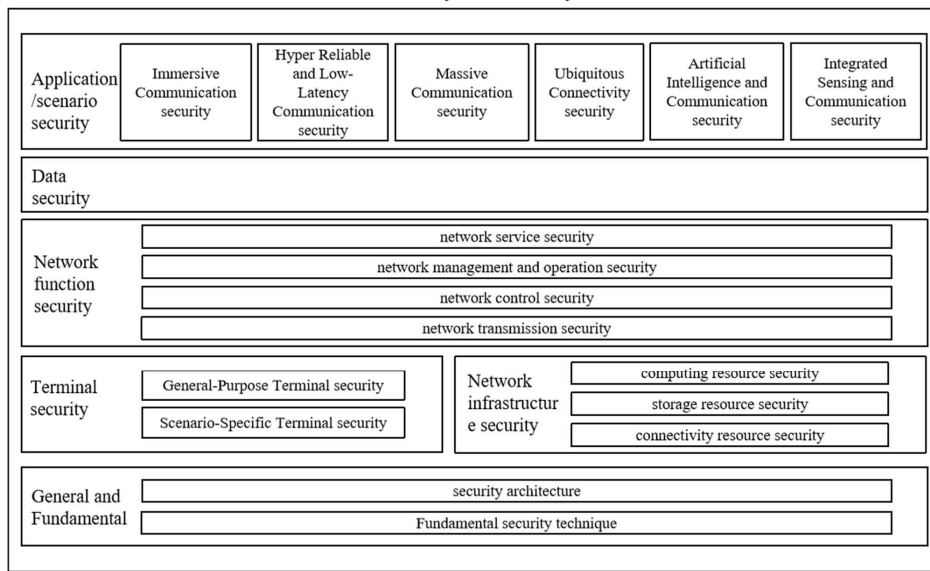
Considering the stages of standard application and implementation, 6G security standards are categorized into security technology standards, security evaluation standards, etc.

1. Security technology standards: These standards refer to specific security specifications that use security technical means to ensure the security of 6G networks and their services.

2. Security evaluation standards: These standards are used to assess the security level of 6G networks, systems, products, and data, covering security level assessment, evaluation index requirements, and evaluation methods.

The IMT 2030 Security Standard System is shown in the figure below.

IMT 2030 Security Standard System



서식 있음: 첫 줄: 0 글자, 글머리 기호 + 수준:1 + 맞춤
위치: 0 cm + 들여쓰기 위치: 0.74 cm

서식 있음: 글머리 기호 및 번호 매기기

서식 있음: 들여쓰기: 첫 줄: 0 글자

Figure 1 – IMT 2030 Security Standard System

The mapping relationship between the security issues identified in Chapter 7 and the standard system is shown in the figure below.

//editor note: update the mapping of the new key issues.

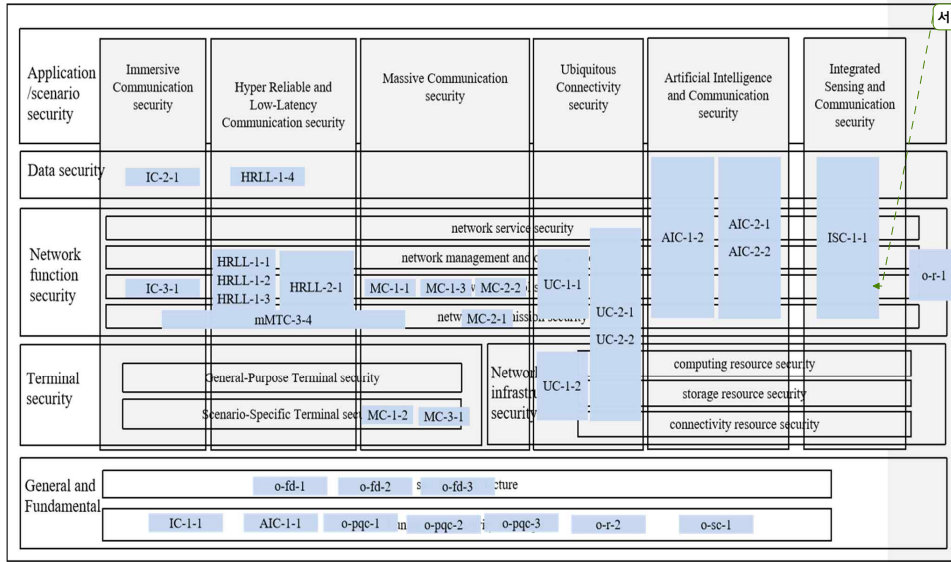


Figure 2 – IMT 2030 Security Standard System for key issues

6G security standards contains refined research on security requirements and solutions, which can be used to address the interoperability and provide implementation guidelines for the aforementioned key issues.

For each category of standards included in the standard system, a list of standards as shown in Table 1.13 has been developed by integrating relevant domestic and international standards that have been released or are under development. This list details the specific standards covered by each category.

This content helps to understand the overall progress of the 6G security standard work, the standards that need to be developed and the main key issues they should address. It also helps to identify the deliverables that can be referenced or those for which gaps need to be analyzed, thereby avoiding problems such as missing standards or duplicate standards and promoting cooperation with other organizations to advance the development of 6G security standards.

Appendix I

Standard list related to IMT-2030 network security

This appendix lists the specific standards and category related to IMT-2030 network security.

서식 있음: 표준, 템: 3.31 글자, 왼쪽 + 4.96 글자, 왼쪽 + 6.62 글자, 왼쪽 + 8.27 글자, 왼쪽

Table 1 – IMT 2030 Security Standard list

Item Number	Subject	Key issue	Progress	Relevant Standards	
0. Roadmap					
1	Overall roadmap on IMT2030 security	All	ITU-T XSTR. IMT2030-sec-con: Security Consideration for IMT-2030 Networks (2026)		
A. Overall Architecture					
2	Overall Architecture on IMT2030 security	0-fd-1, 0-fd-2, 0-fd-3		3GPP FS 6G SEC: Study on Security for the 6G System (2027); ITU X.805 (2008)	
B. Fundamental technique					
3	PQC algorithms	0-pqc-1		NIST	
4	PQC application guideline	0-pqc-2, 0-pqc-3		3GPP FS CryptoPQC: Study on Transitioning to Post Quantum Cryptography in 3GPP (2026)	
5	Security guideline for AI in IMT-2030	AIC-1-1		ITU X.sr-ai: Security requirements for AI systems ()	
C. Security technology standards					
C.1 Application/scenario security					
6	Security guidelines for Immersive Communication	IC-1-1, IC-2-1, IC-3-1			
7	Security guidelines for Hyper Reliable and Low-Latency Communication	HRLL-1-1, HRLL-1-2, HRLL-1-3, HRLL-1-4, HRLL-2-1, HRLL-2-2			
8	Security guidelines for Massive Communication	MC-1-1, MC-1-2, MC-1-3, MC-2-1, MC-2-2			
9	Security guidelines for Passive IoT	MC-3-1			

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10	Security guidelines for Ubiquitous Connectivity	UC-2-1, UC-2-2		GSMA Decentralised trust and security for 6G()
11	Security guidelines for non-terrestrial communication systems	UC-1-1, UC-1-2	ITU TR.FMSC-IMT2030 : Security technologies for fixed, mobile and satellite convergence of IMT-2030 networks (2027)	
12	Security guidelines for Artificial Intelligence and Communication	AIC-1-1		
13	Security guidelines for Integrated Sensing and Communication	ISC-1-1	ITU X.s-isac: Security guidelines for integrated sensing and communication in IMT-2020 networks and beyond (2027)	
C.2 Data security standards				
14	Security guidelines for data of IMT-2030 network			
C.3 network function standards				
15	Security guidelines for services of IMT-2030 network	AIC-1-2, AIC-2-1, AIC-2-2, ISC-1-1		
16	Security guidelines for management and operation of IMT-2030 network	AIC-1-2		
17		AIC-2-1		ITU X.2011:security guidelines for digital twin(2024);
18		AIC-2-2		ITU X.dtns:Guidelines of using digital twin of network for network securit(2026)
19	Security guidelines for control of IMT-2030 network	AIC-1-2, AIC-2-1, AIC-2-2, ISC-1-1		3GPP FS_6G_SEC:Study on Security for the 6G System(2027)
20	Security guidelines for transmission of IMT-2030 network	AIC-1-2, AIC-2-1, AIC-2-2, ISC-1-1		3GPP FS_6G_SEC:Study on Security for the 6G System(2027)
C.4 Terminal security standards				

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<u>21</u>	<u>Security guidelines for general purpose terminal of IMT-2030 network</u>				
<u>22</u>	<u>Security guidelines for specific terminal of IMT-2030 network</u>	<u>MC-1-2, MC-3-1</u>			
<u>23</u>					
C.4 Network infrastructure security standards					
<u>24</u>	<u>Security guidelines for computing resource of IMT-2030 network</u>	<u>UC-1-2, UC-2-1, UC-2-2</u>		<u>ITU TR.sg-lmcs:Security guidelines of DLT-based lifecycle management for computing services(2025);</u> <u>ITU TR.sd-cnc:Security guidelines for data of coordination of networking and computing(2026)</u>	
<u>25</u>	<u>Security guidelines for storage resource of IMT-2030 network</u>	<u>UC-1-2, UC-2-1, UC-2-2</u>			
<u>26</u>	<u>Security guidelines for connectivity resource of IMT-2030 network</u>	<u>UC-1-2, UC-2-1, UC-2-2</u>			
D. Security evaluation standards					
<u>27</u>					

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Appendix II

Some potential key technologies for securing IMT-2030 network

This appendix I provides some potential underling key technologies for securing IMT-2030 network.

II.1 Software supply chain security

As IMT2030/6G networks evolve into highly software-driven, cloud-native, and AI-integrated infrastructures, the security of the software supply chain becomes a critical concern. IMT2030 networks increasingly depend on open-source components, third-party libraries, and complex CI/CD pipelines. This dependency expands the attack surface, making malicious code injection, dependency hijacking, and compromised updates realistic threats.

In the IMT2030/6G context, where ultra-low latency, massive device connectivity, and network slicing are central, any compromise in the software supply chain could cascade rapidly across multiple domains, affecting critical services such as autonomous systems, industrial automation, and public safety communications. Unlike traditional threats, supply chain compromises may bypass conventional perimeter defenses by arriving as “trusted” updates or components. Studying software supply chain security in 6G is therefore essential to:

- develop methods to verify the integrity and provenance of all software components.
- introduce continuous monitoring and attestation mechanisms for deployed workloads.
- establish best practices for secure development, distribution, and maintenance of telecom software.
- enhance resilience by enabling rapid detection, isolation, and remediation of supply chain compromises.

II.2 Zero Trust Technology

The Zero Trust security architecture is based on the principle of “never trust, always verify”. Unlike traditional perimeter-based security, where devices and users inside the network are assumed to be trustworthy, Zero Trust requires continuous authentication, authorization, and validation of every entity, regardless of its location within the network.

In the context of 6G networks, Zero Trust plays a pivotal role in addressing the massive attack surface created by hyper-connectivity, network slicing, and cloud-native infrastructures. With billions of heterogeneous devices—including IoT endpoints, autonomous vehicles, drones, and mission-critical systems—6G cannot rely on static trust boundaries. Instead, Zero Trust ensures that each access request is evaluated dynamically based on identity, device posture, context, and risk level.

The adoption of Zero Trust technologies in 6G provides several key security benefits:

- Enhanced access control: Continuous verification of users, devices, and services reduces the risk of unauthorized access across network slices and edge domains.
- Containment of security breaches: By enforcing micro-segmentation and least-privilege principles, Zero Trust prevents lateral movement of attackers within virtualized or cloud-native infrastructures.
- Adaptive security for dynamic environments: Since 6G networks are highly programmable and service-based, Zero Trust allows security policies to adapt in real time to context changes, such as device mobility, edge reconfiguration, or AI-driven orchestration.

- Resilience against insider and supply chain threats: By removing implicit trust, Zero Trust mitigates risks from compromised internal components or malicious insiders.

Ultimately, Zero Trust provides a foundational security paradigm for 6G networks, ensuring that even in highly distributed, AI-driven, and ultra-dense environments, security is maintained through continuous validation, minimal trust assumptions, and strict enforcement of access controls.

[editor note: need more discussion in the future]

I.3 AI/ML Security & Adversarial Attacks

IMT2030/6G networks are expected to be highly intelligent, software-defined, and data-driven, leveraging AI and ML across network management, resource allocation, anomaly detection, predictive maintenance, and autonomous decision-making. While these capabilities promise unprecedented efficiency and adaptability, they also introduce new security risks. AI and ML models can themselves become targets of attacks that compromise network operations, privacy, and reliability. Adversarial attacks, including data poisoning, model evasion, and inference attacks, can manipulate AI/ML outputs or extract sensitive information from models. In 6G, such attacks could affect critical services like autonomous vehicles, industrial automation, remote healthcare, and smart cities, where incorrect decisions or data leakage could have severe consequences. Studying AI/ML security in the 6G context is essential to:

- develop robust AI/ML models resilient to adversarial manipulation.
- implement secure training and inference pipelines, including data validation, anomaly detection, and privacy-preserving techniques.
- detect, mitigate, and respond to adversarial attacks in real-time, ensuring safe network operations.

establish standards and best practices for trustworthy AI integration in telecom networks.

I.4 Post-Quantum Cryptography

As IMT2030/6G networks are being designed to support ultra-high-speed connectivity, massive device density, and critical infrastructure applications, the security of communications becomes paramount. Traditional public-key cryptographic schemes, such as RSA and ECC, rely on mathematical problems that are vulnerable to quantum computing attacks. With the anticipated emergence of practical quantum computers, these widely used algorithms may be broken, potentially compromising confidentiality, integrity, and authentication across telecom networks.

IMT2030/6G introduces new paradigms such as network slicing, edge computing, AI-driven services, and massive IoT deployments, all of which depend on strong cryptographic guarantees. A single vulnerability in key exchange, digital signatures, or secure boot processes could be catastrophic when propagated across billions of connected devices and critical services.

Studying Post-Quantum Cryptography (PQC) in the IMT2030/6G context is essential to:

- Identify and adopt quantum-resistant algorithms for key exchange, digital signatures, and encryption.
- Ensure secure provisioning and lifecycle management of devices and network functions in a post-quantum era.
- Integrate PQC into high-performance, low-latency telecom environments, balancing security with efficiency.

Prepare standards and implementation guidelines to future-proof IMT2030/6G networks against quantum-enabled attacks.

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